

6502 Machine Language Opcodes (Alphabetical Order – in hexadecimal)

Opcode	Definition	Immediate	Accumulator	Zero Page	Zero Page,X	Zero Page,Y	Absolute	Absolute,X	Absolute,Y	Indirect Abs.	(Zero Page,X)	(Zero Page),Y	Implied	Relative	Affects Flags
	Abbreviations	imm	acc	zp	zp,x	zp,y	abs	abs,x	abs,y	ind	(zp,x)	(zp),y	imp	rel	
ADC	Add with Carry	69		65	75		6D	7D	79		61	71			N,V,Z,C
AND	Bitwise AND with Accumulator	29		25	35		2D	3D	39		21	31			N,Z
ASL	Arithmetic Shift Left		0A	06	16		0E	1E							N,Z,C
BCC	Branch on Carry Clear													90	
BCS	Branch on Carry Set													B0	
BEQ	Branch on Equal													F0	
BIT	Test Bits			24			2C								N,V,Z
BMI	Branch on Minus													30	
BNE	Branch on Not Equal													D0	
BPL	Branch on Plus													10	
BRK	Break												00		B
BVC	Branch on Overflow Clear													50	
BVS	Branch on Overflow Set													70	
CLC	Clear Carry												18		C
CLD	Clear Decimal												D8		D
CLI	Clear Interrupt												58		I
CLV	Clear Overflow												B8		V
CMP	Compare Accumulator	C9		C5	D5		CD	DD	D9		C1	D1			N,Z,C
CPX	Compare X Register	E0		E4			EC								N,Z,C
CPY	Compare Y Register	C0		C4			CC								N,Z,C
DEC	Decrement Memory			C6	D6		CE	DE							N,Z
DEX	Decrement X												CA		N,Z
DEY	Decrement Y												88		N,Z
EOR	Bitwise Exclusive OR	49		45	55		4D	5D	59		41	51			N,Z
INC	Increment Memory			E6	F6		EE	FE							N,Z
INX	Increment X												E8		N,Z
INY	Increment Y												C8		N,Z
JMP	Jump						4C			6C					
JSR	Jump to Subroutine						20								
LDA	Load Accumulator	A9		A5	B5		AD	BD	B9		A1	B1			N,Z
LDX	Load X Register	A2		A6		B6	AE		BE						N,Z
LDY	Load Y Register	A0		A4	B4		AC	BC							N,Z
LSR	Logical Shift Right		4A	46	56		4E	5E							N,Z,C
NOP	No Operation												EA		
ORA	Bitwise OR with Accumulator	09		05	15		0D	1D	19		01	11			N,Z
PHA	Push Accumulator to Stack												48		
PHP	Push Processor Status to Stack												08		
PLA	Pull Accumulator off Stack												68		N,Z
PLP	Pull Processor Status off Stack												28		ALL
ROL	Rotate Left		2A	26	36		2E	3E							N,Z,C
ROR	Rotate Right		6A	66	76		6E	7E							N,Z,C
RTI	Return from Interrupt												40		ALL
RTS	Return from Subroutine												60		
SBC	Subtract with Carry	E9		E5	F5		ED	FD	F9		E1	F1			N,V,Z,C
SEC	Set Carry												38		C
SED	Set Decimal												F8		D
SEI	Set Interrupt												78		I
STA	Store Accumulator			85	95		8D	9D	99		81	91			
STX	Store X Register			86		96	8E								
STY	Store Y Register			84	94		8C								
TAX	Transfer A to X												AA		N,Z
TAY	Transfer A to Y												A8		N,Z
TSX	Transfer Stack Pointer to X												BA		
TXA	Transfer X to A												8A		N,Z
TXS	Transfer X to Stack Pointer												9A		
TYA	Transfer Y to A												98		N,Z

The result of the opcode directly modifies....

System Memory (Gray 2)
8-bit Registers (Accumulator/X/Y) (Sky blue 10)
16-bit Registers (AB, CD, EF, etc.) (Magenta 10)
CPU Status Register (Orange 1)
Stack/Stack Pointer (Red 2)
Program Counter (Green 2)

Flags

N	Negative (1 = negative)
V	Overflow (1 = true)
B	BRK Instruction (1 = BRK, 0 = IRQB)
D	Decimal Mode (1 = true)
I	IRQB Disable (1 = disable)
Z	Zero Result (1 = true)
C	Carry (1 = true)

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This table originally authored by Lawrence Woodman (https://github.com/lawrencewoodman/machine_language_aids)

Opcode color coding and formatting changes by Brian Manning (https://github.com/spicyjack/8-16_bit_cpu_info)

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6502 Machine Language Opcodes (By Opcode Group - in hexadecimal)

Opcode	Definition	Immediate	Accumulator	Zero Page	Zero Page,X	Zero Page,Y	Absolute	Absolute,X	Absolute,Y	Indirect Abs.	(Zero Page,X)	(Zero Page),Y	Implied	Relative	Affects Flags
	Abbreviations	imm	acc	zp	zp,x	zp,y	abs	abs,x	abs,y	ind.abs	(zp,x)	(zp),y	imp	rel	
Load/Store/Transfer															
LDA	Load Accumulator	A9		A5	B5		AD	BD	B9		A1	B1			N,Z
LDX	Load X Register	A2		A6		B6	AE		BE						N,Z
LDY	Load Y Register	A0		A4	B4		AC	BC							N,Z
STA	Store Accumulator			85	95		8D	9D	99		81	91			
STX	Store X Register			86		96	8E								
STY	Store Y Register			84	94		8C								
TAX	Transfer A to X												AA		N,Z
TAY	Transfer A to Y												A8		N,Z
TSX	Transfer Stack Pointer to X												BA		
TXA	Transfer X to A												8A		N,Z
TXS	Transfer X to Stack Pointer												9A		
TYA	Transfer Y to A												98		N,Z
Arithmetic/Logic															
ADC	Add with Carry	69		65	75		6D	7D	79		61	71			N,V,Z,C
CMP	Compare Accumulator	C9		C5	D5		CD	DD	D9		C1	D1			N,Z,C
CPX	Compare X Register	E0		E4			EC								N,Z,C
CPY	Compare Y Register	C0		C4			CC								N,Z,C
SBC	Subtract with Carry	E9		E5	F5		ED	FD	F9		E1	F1			N,V,Z,C
Bit Manipulation/Rotate/Shift															
AND	Bitwise AND with Accumulator	29		25	35		2D	3D	39		21	31			N,Z
ASL	Arithmetic Shift Left		0A	06	16		0E	1E							N,Z,C
BIT	Test Bits			24			2C								N,V,Z
EOR	Bitwise Exclusive OR	49		45	55		4D	5D	59		41	51			N,Z
LSR	Logical Shift Right		4A	46	56		4E	5E							N,Z,C
ORA	Bitwise OR with Accumulator	09		05	15		0D	1D	19		01	11			N,Z
ROL	Rotate Left		2A	26	36		2E	3E							N,Z,C
ROR	Rotate Right		6A	66	76		6E	7E							N,Z,C
Increment/Decrement/Loops/Indexes/Offsets															
DEC	Decrement Memory			C6	D6		CE	DE							N,Z
DEX	Decrement X												CA		N,Z
DEY	Decrement Y												88		N,Z
INC	Increment Memory			E6	F6		EE	FE							N,Z
INX	Increment X												E8		N,Z
INY	Increment Y												C8		N,Z
Jumping/Branching/Returns															
BCC	Branch on Carry Clear													90	
BCS	Branch on Carry Set													B0	
BEQ	Branch on Equal													F0	
BMI	Branch on Minus													30	
BNE	Branch on Not Equal													D0	
BPL	Branch on Plus													10	
BVC	Branch on Overflow Clear													50	
BVS	Branch on Overflow Set													70	
JMP	Jump						4C			6C					
JSR	Jump to Subroutine						20								
RTI	Return from Interrupt												40		ALL
RTS	Return from Subroutine												60		
Stack Ops															
PHA	Push Accumulator to Stack												48		
PHP	Push Processor Status to Stack												08		
PLA	Pull Accumulator off Stack												68		N,Z
PLP	Pull Processor Status off Stack												28		ALL
CPU Control (Break, NOP, Set/Clear Flags)															
BRK	Break												00		B
CLC	Clear Carry												18		C
CLD	Clear Decimal												D8		D
CLI	Clear Interrupt												58		I
CLV	Clear Overflow												B8		V
NOP	No Operation												EA		
SEC	Set Carry												38		C
SED	Set Decimal												F8		D
SEI	Set Interrupt												78		I

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Program Counter (Green 2)

Flags

N Negative (1 = negative)

V Overflow (1 = true)

B BRK Instruction (1 = BRK, 0 = IRQB)

D Decimal Mode (1 = true)

I IRQB Disable (1 = disable)

Z Zero Result (1 = true)

C Carry (1 = true)

Hex	Instruction	ASCII	Dec	Oct	BinH	BinL	Hex	Instruction	ASCII	Dec	Oct	BinH	BinL
00	BRK	NUL	0	000	0000	0000	40	RTI	@	64	100	0100	0000
01	ORA (zp,x)	SOH	1	001	0000	0001	41	EOR (zp,x)	A	65	101	0100	0001
02	-	STX	2	002	0000	0010	42	-	B	66	102	0100	0010
03	-	ETX	3	003	0000	0011	43	-	C	67	103	0100	0011
04	-	EOT	4	004	0000	0100	44	-	D	68	104	0100	0100
05	ORA zp	ENQ	5	005	0000	0101	45	EOR zp	E	69	105	0100	0101
06	ASL zp	ACK	6	006	0000	0110	46	LSR zp	F	70	106	0100	0110
07	-	BEL	7	007	0000	0111	47	-	G	71	107	0100	0111
08	PHP	BS	8	010	0000	1000	48	PHA	H	72	110	0100	1000
09	ORA imm	HT	9	011	0000	1001	49	EOR imm	I	73	111	0100	1001
0A	ASL acc	LF	10	012	0000	1010	4A	LSR acc	J	74	112	0100	1010
0B	-	VT	11	013	0000	1011	4B	-	K	75	113	0100	1011
0C	-	FF	12	014	0000	1100	4C	JMP abs	L	76	114	0100	1100
0D	ORA abs	CR	13	015	0000	1101	4D	EOR abs	M	77	115	0100	1101
0E	ASL abs	SO	14	016	0000	1110	4E	LSR abs	N	78	116	0100	1110
0F	-	SI	15	017	0000	1111	4F	-	O	79	117	0100	1111
10	BPL	DLE	16	020	0001	0000	50	BVC	P	80	120	0100	0000
11	ORA (zp),y	DC1	17	021	0001	0001	51	EOR (zp),y	Q	81	121	0101	0001
12	-	DC2	18	022	0001	0010	52	-	R	82	122	0101	0010
13	-	DC3	19	023	0001	0011	53	-	S	83	123	0101	0011
14	-	DC4	20	024	0001	0100	54	-	T	84	124	0101	0100
15	ORA zp,x	NAK	21	025	0001	0101	55	EOR zp,x	U	85	125	0101	0101
16	ASL zp,x	SYN	22	026	0001	0110	56	LSR zp,x	V	86	126	0101	0110
17	-	ETB	23	027	0001	0111	57	-	W	87	127	0101	0111
18	CLC	CAN	24	030	0001	1000	58	CLI	X	88	130	0101	1000
19	ORA abs,y	EM	25	031	0001	1001	59	EOR abs,y	Y	89	131	0101	1001
1A	-	SUB	26	032	0001	1010	5A	-	Z	90	132	0101	1010
1B	-	ESC	27	033	0001	1011	5B	-	[	91	133	0101	1011
1C	-	FS	28	034	0001	1100	5C	-	\	92	134	0101	1100
1D	ORA abs,x	GS	29	035	0001	1101	5D	EOR abs,x	]	93	135	0101	1101
1E	ASL abs,x	RS	30	036	0001	1110	5E	LSR abs,x	^	94	136	0101	1110
1F	-	US	31	037	0001	1111	5F	-	~	95	137	0101	1111
20	JSR (Space)		32	040	0010	0000	60	RTS	·	96	140	0110	0000
21	AND (zp,x)	!	33	041	0010	0001	61	ADC (zp,x)	a	97	141	0110	0001
22	-	"	34	042	0010	0010	62	-	b	98	142	0110	0010
23	-	#	35	043	0010	0011	63	-	c	99	143	0110	0011
24	BIT zp	\$	36	044	0010	0100	64	-	d	100	144	0110	0100
25	AND zp	%	37	045	0010	0101	65	ADC zp	e	101	145	0110	0101
26	ROL zp	&	38	046	0010	0110	66	ROR zp	f	102	146	0110	0110
27	-	'	39	047	0010	0111	67	-	g	103	147	0110	0111
28	PLP	(	40	050	0010	1000	68	PLA	h	104	150	0110	1000
29	AND imm	)	41	051	0010	1001	69	ADC imm	i	105	151	0110	1001
2A	ROL acc	*	42	052	0010	1010	6A	ROR acc	j	106	152	0110	1010
2B	-	+	43	053	0010	1011	6B	-	k	107	153	0110	1011
2C	BIT abs	,	44	054	0010	1100	6C	JMP ind	l	108	154	0110	1100
2D	AND abs	-	45	055	0010	1101	6D	ADC abs	m	109	155	0110	1101
2E	ROL abs	.	46	056	0010	1110	6E	ROR abs	n	110	156	0110	1110
2F	-	/	47	057	0010	1111	6F	-	o	111	157	0110	1111
30	BMI	0	48	060	0011	0000	70	BVS	p	112	160	0111	0000
31	AND (zp),y	1	49	061	0011	0001	71	ADC (zp),y	q	113	161	0111	0001
32	-	2	50	062	0011	0010	72	-	r	114	162	0111	0010
33	-	3	51	063	0011	0011	73	-	s	115	163	0111	0011
34	-	4	52	064	0011	0100	74	-	t	116	164	0111	0100
35	AND zp,x	5	53	065	0011	0101	75	ADC zp,x	u	117	165	0111	0101
36	ROL zp,x	6	54	066	0011	0110	76	ROR zp,x	v	118	166	0111	0110
37	-	7	55	067	0011	0111	77	-	w	119	167	0111	0111
38	SEC	8	56	070	0011	1000	78	SEI	x	120	170	0111	1000
39	AND abs,y	9	57	071	0011	1001	79	ADC abs,y	y	121	171	0111	1001
3A	-	:	58	072	0011	1010	7A	-	z	122	172	0111	1010
3B	-	;	59	073	0011	1011	7B	-	{	123	173	0111	1011
3C	-	<	60	074	0011	1100	7C	-		124	174	0111	1100
3D	AND abs,x	=	61	075	0011	1101	7D	ADC abs,x	}	125	175	0111	1101
3E	ROL abs,x	>	62	076	0011	1110	7E	ROR abs,x	~	126	176	0111	1110
3F	-	?	63	077	0011	1111	7F	-	DEL	127	177	0111	1111

Hex	Instruction	ASCII	Dec	Oct	BinH	BinL	Hex	Instruction	ASCII	Dec	Oct	BinH	BinL
80	-		128	000	1000	0000	C0	CPY imm		192	100	1100	0000
81	STA (zp,x)		129	001	1000	0001	C1	CMP (zp,x)		193	101	1100	0001
82	-		130	002	1000	0010	C2	-		194	102	1100	0010
83	-		131	003	1000	0011	C3	-		195	103	1100	0011
84	STY zp		132	004	1000	0100	C4	CPY zp		196	104	1100	0100
85	STA zp		133	005	1000	0101	C5	CMP zp		197	105	1100	0101
86	STX zp		134	006	1000	0110	C6	DEC zp		198	106	1100	0110
87	-		135	007	1000	0111	C7	-		199	107	1100	0111
88	DEY imp		136	010	1000	1000	C8	INY imp		200	110	1100	1000
89	-		137	011	1000	1001	C9	CMP imm		201	111	1100	1001
8A	TXA imp		138	012	1000	1010	CA	DEX imp		202	112	1100	1010
8B	-		139	013	1000	1011	CB	-		203	113	1100	1011
8C	STY abs		140	014	1000	1100	CC	CPY abs		204	114	1100	1100
8D	STA abs		141	015	1000	1101	CD	CMP abs		205	115	1100	1101
8E	STX abs		142	016	1000	1110	CE	DEC abs		206	116	1100	1110
8F	-		143	017	1000	1111	CF	-		207	117	1100	1111
90	BCC		144	020	1001	0000	D0	BNE		208	120	1101	0000
91	STA (zp),y		145	021	1001	0001	D1	CMP (zp),y		209	121	1101	0001
92	-		146	022	1001	0010	D2	-		210	122	1101	0010
93	-		147	023	1001	0011	D3	-		211	123	1101	0011
94	STY zp,x		148	024	1001	0100	D4	-		212	124	1101	0100
95	STA zp,x		149	025	1001	0101	D5	CMP zp,x		213	125	1101	0101
96	STX zp,y		150	026	1001	0110	D6	DEC zp,x		214	126	1101	0110
97	-		151	027	1001	0111	D7	-		215	127	1101	0111
98	TYA imp		152	030	1001	1000	D8	CLD		216	130	1101	1000
99	STA abs,y		153	031	1001	1001	D9	CMP abs,y		217	131	1101	1001
9A	TXS imp		154	032	1001	1010	DA	-		218	132	1101	1010
9B	-		155	033	1001	1011	DB	-		219	133	1101	1011
9C	-		156	034	1001	1100	DC	-		220	134	1101	1100
9D	STA abs,x		157	035	1001	1101	DD	CMP abs,x		221	135	1101	1101
9E	-		158	036	1001	1110	DE	DEC abs,x		222	136	1101	1110
9F	-		159	037	1001	1111	DF	-		223	137	1101	1111
A0	LDY imm		160	040	1010	0000	E0	CPX imm		224	140	1110	0000
A1	LDA (zp,x)		161	041	1010	0001	E1	SBC (ind,x)		225	141	1110	0001
A2	LDX imm		162	042	1010	0010	E2	-		226	142	1110	0010
A3	-		163	043	1010	0011	E3	-		227	143	1110	0011
A4	LDY zp		164	044	1010	0100	E4	CPX zp		228	144	1110	0100
A5	LDA zp		165	045	1010	0101	E5	SBC zp		229	145	1110	0101
A6	LDX zp		166	046	1010	0110	E6	INC zp		230	146	1110	0110
A7	-		167	047	1010	0111	E7	-		231	147	1110	0111
A8	TAY imp		168	050	1010	1000	E8	INX imp		232	150	1110	1000
A9	LDA imm		169	051	1010	1001	E9	SBC imm		233	151	1110	1001
AA	TAX imp		170	052	1010	1010	EA	NOP		234	152	1110	1010
AB	-		171	053	1010	1011	EB	-		235	153	1110	1011
AC	LDY abs		172	054	1010	1100	EC	CPX abs		236	154	1110	1100
AD	LDA abs		173	055	1010	1101	ED	SBC abs		237	155	1110	1101
AE	LDX abs		174	056	1010	1110	EE	INC abs		238	156	1110	1110
AF	-		175	057	1010	1111	EF	-		239	157	1110	1111
B0	BCS		176	060	1011	0000	F0	BEQ		240	160	1111	0000
B1	LDA (zp),y		177	061	1011	0001	F1	SBC (zp),y		241	161	1111	0001
B2	-		178	062	1011	0010	F2	-		242	162	1111	0010
B3	-		179	063	1011	0011	F3	-		243	163	1111	0011
B4	LDY zp,x		180	064	1011	0100	F4	-		244	164	1111	0100
B5	LDA zp,x		181	065	1011	0101	F5	SBC zp,x		245	165	1111	0101
B6	LDX zp,y		182	066	1011	0110	F6	INC zp,x		246	166	1111	0110
B7	-		183	067	1011	0111	F7	-		247	167	1111	0111
B8	CLV		184	070	1011	1000	F8	SED		248	170	1111	1000
B9	LDA abs,y		185	071	1011	1001	F9	SBC abs,y		249	171	1111	1001
BA	TSX imp		186	072	1011	1010	FA	-		250	172	1111	1010
BB	-		187	073	1011	1011	FB	-		251	173	1111	1011
BC	LDY abs,x		188	074	1011	1100	FC	-		252	174	1111	1100
BD	LDA abs,x		189	075	1011	1101	FD	SBC abs,x		253	175	1111	1101
BE	LDX abs,y		190	076	1011	1110	FE	INC abs,x		254	176	1111	1110
BF	-		191	077	1011	1111	FF	-		255	177	1111	1111

"ASCII" has no "official" character codes above 0x7F/127; different platforms place platform-specific characters in this area.

Some suggestions for characters for values starting at 0x80/128:

<https://en.wikipedia.org/wiki/Windows-1252>

https://en.wikipedia.org/wiki/ISO/IEC_8859-1

		Second Hex Digit															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
First Hex Digit	BRK																
	0x0																
	0x1																
	0x2																
	0x3																
	0x4																
	0x5																
	0x6																
	0x7																
	0x8																
	0x9																
	0xA																
	0xB																
	0xC																
	0xD																
	0xE																
	0xF																

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- System Memory (Gray 2)
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- 16-bit Registers (AB, CD, EF, etc.) (Magenta 10)
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